

Microsoft Foundry Local Lab: Portfolio & Project Index

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Environment Path: C:\Users\cbern\OneDrive\Documents\Feb 2026\CM on AI 2026\Foundry-Local-Lab\python
System Configuration: Local LLM Inference (Phi-3.5-mini), SQLite Vector/TF-IDF Database, Streamlit Web UI, and Zero-Trust Public Tunneling (zrok).

1. Executive Summary & Portfolio Objective

This document serves as a structured registry, code index, and architectural "org chart" of coursework and custom development completed during the Microsoft Foundry Local crash course. The goal of this repository is to pivot foundational local LLM capabilities (inference, context injection, function calling) into production-grade agentic tools for commercial underwriting, actuarial analytics, and reinsurance market intelligence. By repurposing core lesson templates into specialized scripts, this index acts as a technical portfolio demonstrating mastery over local, secure AI engineering without reliance on cloud-monolithic setups.

2. Curriculum Index & Project Registry

The following index maps the structural progression of the Foundry-Local-Lab\python directory. Each subfolder starting with the lowercase cm prefix denotes a targeted implementation or repurposed extension of a core framework lesson.

Module / Folder	Core Topic	Repurposed Scripts & Components	Technical Synopsis
cm lesson 1 - 2	Local Inference & SDK Integration	inference_test.py client_init.py	Establishing stable connections to the local Microsoft Foundry service endpoint. Configured local orchestration layer to offload processing, verify NPU/GPU engagement parameters, and

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			handle streaming chat completions cleanly.
cm lesson 3 - 5	Context Control & Token Management	token_counter.py system_prompt_builder.py	Optimizing token parameters to avoid response truncation during long-form technical generation. Set up custom systemic rules forcing structured, highly analytical insurance and actuarial outputs using a 2,000 max-token allocation.
cm lesson 6 - 8	Structured Output & Schema Enforcement	schema_validator.py json_parser_local.py	Enforcing JSON-schema constraints directly on local models. Designed to parse unstructured casualty treaty text or slip details into rigid data structures without breaking syntax constraints.
cm lesson 9 - 10	Local Knowledge Bases & SQLite RAG	RAG_TF-IDF_SQLite.py	The foundational data engine. Implements a localized Retrieval-Augmented Generation layout. Parses market documents (e.g., Swiss Re, Gen Re

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			reports), generates TF-IDF vectors, maintains an active file cache index in reinsurance_kb.db, and executes source scoring and citation mechanisms.
cm lesson 11	User Interface & Broker Models	RAG_Web_Portal.py	A local dashboard interface leveraging Streamlit. Features a secure "Knowledge Management" sidebar for monitoring indexed documents, source scoreboard tracking, custom CSS styling injection, and dynamic query input routing to the underlying SQLite database engine.
cm lesson 12+	Agentic Architectures & Custom Writers	repurposed_zaza_writer.py agent_handoff_core.py	Transitioning from single LLM execution to collaborative, tool-enabled agents. Built around the ZazaCreativeWriter template to explore multi-layered technical drafting (e.g., market commentary, underwriting reasoning updates). Identified as a

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			primary deep-dive target for non-monolithic multi-agent expansion.

Note: Additional customized personal utilities (e.g., 1A Compendium) are slated for integration into this registry following data normalization.

3. Core Architectural Highlights

- **The SQLite RAG Core (RAG_TF-IDF_SQLite.py):** Implements hard-filtering pre-checks (date and keyword vector alignments) against localized reference libraries to significantly curb hallucination risks common to compact local models.
- **Web-Enabled Distribution (Streamlit + zrok):** Combines a clean frontend presentation layer with zero-trust tunneling infrastructure. By binding local Streamlit instances (port 8501) via persistent, basic-auth secured zrok reserved proxies, the architecture acts as an enterprise-grade "Broker" prototype where data processing remains local while delivery remains flexible.
- **The Agentic Pivot:** Focuses explicitly on distinct, non-monolithic agentic interaction layers rather than highly integrated, inflexible single-block routines, laying the groundwork for custom underwriting multi-agent frameworks.